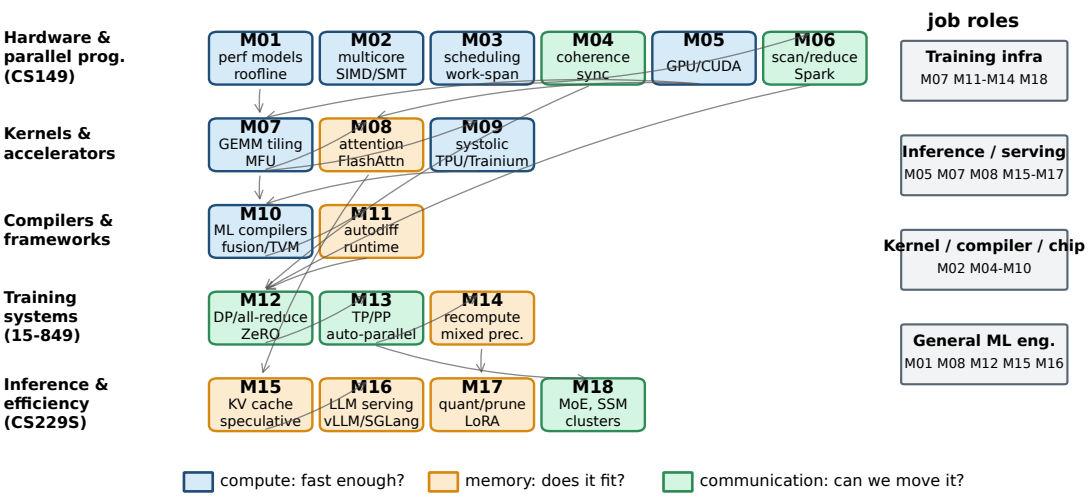
Hardware & parallel prog. (CS149)
Kernels & accelerators
Compilers & frameworks
Training systems (15-849)
Inference & efficiency (CS229S)
M01 perf models roofline
M02 multicore SIMD/SMT
M03 scheduling work-span
M04 coherence sync
M05 GPU/CUDA
M06 scan/reduce Spark
M07 GEMM tiling MFU
M08 attention FlashAttn
M09 systolic TPU/Trainium
M10 ML compilers fusion/TVM
M11 autodiff runtime
M12 DP/all-reduce ZeRO
M13 TP/PP auto-parallel
M14 recompute mixed prec.
M15 KV cache speculative
M16 LLM serving vLLM/SGLang
M17 quant/prune LoRA
M18 MoE, SSM clusters
job roles
Training infra
M07 M11-M14 M18
Inference / serving
M05 M07 M08 M15-M17
Kernel / compiler / chip
M02 M04-M10
General ML eng.
M01 M08 M12 M15 M16
compute: fast enough?
memory: does it fit?
communication: can we move it?